JanaМен full-project QA report — 17 October 2026

Overall project readiness: **NEEDS WORK**. The local full-stack MVP and MOCK gateway have substantial executed coverage; remaining operational, privacy-governance, duplicate-submission and accessibility gaps prevent a production-readiness claim.

Checklist counts (162 requirements): PASS: 140 / FAIL: 2 / PARTIAL: 17 / NOT VERIFIED: 2 / BLOCKED: 0 / NOT APPLICABLE: 1. Findings use separate severity/status labels; their count is not added to requirement totals. All 16 required acceptance scenarios have executed test evidence.

Two P1 defects were reproduced and fixed: conflicting literal numeric policy claims being returned as grounded, and raw employee questions carrying synthetic identifiers into the mocked Gemini HTTP request. No live provider was called. No additional P0/P1 authorization defect was confirmed in the reviewed paths.

The patched review stack runs at `http://127.0.0.1:18766`. The pre-existing stack on port8000 was not rebuilt; it retains its earlier image until an operator explicitly updates it. These findings describe the saved source and the isolated QA environment.

Executed verification

- Final full SQLite suite: **400 passed**, 1 warning; final full PostgreSQL 17 suite: **400 passed**, 1 warning. Same 400 tests, two database environments; migrations and API end-to-end tests are included.
- **5 frontend locale tests passed**; JavaScript syntax, Ruff check/format, mypy: 76 application sources, pip dependency consistency passed. OSV checked 50 pinned runtime/dev/optional-ML packages and returned 0 advisory matches.
- Actual Docker build/start and stopped-gateway/recovery journey passed. Four services run, three expose configured healthy probes; gateway is internal and has no host port. Approval remains Approved, snapshot loads offline, and recovery yields one MOCK registration/notice.
- Browser inspected employee pages, manager Team and HR card/source/refresh controls. Actual course progress survives process rebuild and HTTP re-login. Keyboard modal defect reproduced; mobile and exhaustive browser coverage remain unverified.

Exact commands, exits, interim failures and environment scope: QA_TEST_RESULTS.md. Per-item assessment: QA_CHECKLIST.md. Prioritized remediation: FIX_PLAN.md.

Five principal remaining risks

1. **External data governance and enterprise deployment are unverified.** Raw questions now stay local, but approved knowledge can itself contain private material. Operator approval must cover external source transmission. Real 1C/Documentolog, production inbound mTLS, backups and model availability have no verified connection/test here.
2. **Operational recovery is incomplete.** HR can inspect failures/mappings; safe authorized replay/remapping is absent. The gateway uses an initial create_all schema and needs versioned upgrades before future schema changes.
3. **Initial advance submission lacks retry idempotency.** Two identical HTTP submissions create two business requests; current request/document idempotency covers later stages. No payroll payment is implemented or claimed.
4. **AI source consistency remains limited.** The fixed numeric pattern is conservative and literal.

Different wording or nonnumeric contradictions require approved policy precedence and further acceptance work.
5. **UX and release confidence have gaps.** Dialog focus escapes, smoke reruns fail on fixed dates, some locale/provenance states are incomplete, and CI/mobile/independent-worker operational coverage is missing.

Repository map and available commands | Area | Real paths / entry points | Verification / startup |
|---|---|---|
| Backend | app/main.py; routers, schemas, services under app/ | `.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000` after explicit migration of the chosen development DB |
| Frontend | static/index.html, static/js/app.js, static/js/i18n.js, static/css/style.css, static/locales/catalog.json | Static HTML/CSS/JS served by FastAPI; `node --check static/js/app.js`; `node --test tests/frontend_localization.test.cjs`; no bundler/build command |
| Database | app/database.py, app/models/__init__.py, app/migrate.py; Alembic 0001..0006 | PostgreSQL 17 in Compose; SQLite supported. `.venv/bin/python -m app.migrate` only against an explicitly selected development/disposable DB |
| Authentication | app/services/auth_service.py, app/routers/auth.py, app/models/session.py | Revocable DB-backed bearer sessions, DB role authority and active-user check; password hashing/legacy upgrade |
| Requests / notifications | app/services/request_service.py, app/services/notification_service.py, app/services/audit_service.py | Transactional request/balance/event/notification/audit/outbox writes; own/team/HR scopes |
| Gateway and worker | app/gateway/main.py, app/gateway/runtime.py, app/integrations/outbox.py, app/integrations/profile_sync.py | Separate `app.gateway.main:app`; `.venv/bin/python -m app.integrations.outbox`; durable jobs, leases, nonces and snapshot interval |
| AI / knowledge | app/services/ai_service.py, app/services/knowledge_conflicts.py, app/integrations/gemini.py, knowledge/README.md, ml/README.md | Offline lexical/TF-IDF with optional learned reranking; optional approved source-only Gemini quote selection; EN/RU/KK backend, RU/KK UI |
| Tests / tooling | pyproject.toml, tests/test_api_matrix.py, tests/test_e2e_journey.py, tests/test_gateway_service.py, tests/test_qa_acceptance.py | pytest; Ruff; mypy; pip check; dependency_audit.py; no tracked CI workflow |
| Deployment/config | Dockerfile, docker-compose.yml, .env.example, docs/INTEGRATIONS.md | Explicit MOCK profile, internal gateway network, optional URL switch; generated private mode0600 config |

The API operation matrix checks every exported OpenAPI operation for authorized behavior, anonymous/invalid-session denial and administrative role restrictions. It does not imply a complete browser or production penetration test. Discovery used the codebase graph first; stale static-JavaScript snippet ranges were checked against actual source.

Documentolog flow traced in current source | Step | Actual implementation | Evidence |
|---|---|---|
| Manager approval / same-transaction business decision |
[app/services/request_service.py:review_request](app/services/request_service.py#L385) |
Conditional Pending transition, frozen debit, history, persistent employee notification, audit and outbox; 3 injected rollback stages tested. |
| Outbox snapshot | [app/integrations/outbox.py:enqueue](app/integrations/outbox.py#L50) |
Immutable canonical payload and stable request:v1 key inserted before approval commit. |
| Worker claim / retries |
[app/integrations/outbox.py:dispatch_one](app/integrations/outbox.py#L242) | PostgreSQL SKIP LOCKED; signed POST or status reconciliation, validated acknowledgement, bounded jittered retries and visible terminal failures. |
| Signing / transport |
[app/integrations/outbox.py:gateway_request](app/integrations/outbox.py#L94) | Existing X-Key-Id/X-Timestamp/X-Nonce/X-Signature helper; raw-body hash and method/path/query binding; no vendor API guessed. |
| Gateway authentication | [app/gateway/main.py:authenticate](app/gateway/main.py#L66) |
Machine signature and durable nonce for every route, independent of frontend bearer roles. |
| Asynchronous durable job | [app/gateway/runtime.py:submit](app/gateway/runtime.py#L92) | POST /v1/documents stores unique idempotent job and returns 202 without synchronous registration. |
| Gateway processing | [app/gateway/runtime.py:process_one](app/gateway/runtime.py#L124) |
Atomic lease claim, durable attempts/recovery; connector and terminal result commit together. |
| MOCK connector | [app/gateway/connectors.py:register](app/gateway/connectors.py#L26) | Unique persisted MOCK document; deterministic reference for the same durable job; controlled rejection/temporary failure modes. |
| Callback sender | [app/gateway/runtime.py:callback_one](app/gateway/runtime.py#L219) |
Persistent callback retries, fresh signed nonce; application GET reconciliation can recover lost callbacks. |
| Signed callback receiver |
[app/routers/internal.py:documentolog_status](app/routers/internal.py#L47) | Signature/body nonce validation, durable replay protection, canonical terminal result validation. |
| Atomic terminal application |
[app/integrations/outbox.py:apply_result](app/integrations/outbox.py#L131) | Matching approved request/job/key, conditional idempotent state update, external reference, one history/audit/registration notice. Business Approved stays separate. |

Source snapshot flow is app/integrations/profile_sync.py → signed MOCK employee-code snapshots → existing profile only → persisted source/as-of/sync metadata; unknown mappings/balance conflicts enter HR review. Profiles/cards read local persisted state and do not synchronously depend on gateway uptime.

Findings and implemented changes

QA-F01 — P1 — FAIL → PASS for reproduced case

Conflicting literal numeric policy claims were labelled grounded.

Evidence: All EN/RU/KK regressions failed before fix; all pass after the guard. All eligible authorized current-language passages are checked before the two-source limit and before Gemini. Source: app/services/knowledge_conflicts.py — `has_conflicting_numeric_claims`.

Fix/result: Implemented; 6 conflict regressions plus equivalence/complement/access tests. General semantic contradiction detection remains a separate P2 gap.

QA-F02 — P1 — FAIL → PASS for reproduced egress

Raw employee free text containing synthetic identifiers reached Gemini HTTP JSON.

Evidence: Three HTTP-boundary regressions failed before fix; they pass after removing raw-question transport and requiring approval both in enabled() and the transport adapter. Source: app/integrations/gemini.py — `generate_grounded_answer`.

Fix/result: Implemented; explicit default-off operator approval, source-only quote selection, retained local coverage/quote verification. Does not scrub PII already placed in approved knowledge.

QA-F03 — P2 — PARTIAL External knowledge data policy and production connector/TLS verification are missing.

Evidence: Approval flag is an operator assertion, not proof of organizational approval. MOCK-only connectors/internal HTTP are intentional; no live vendor or mTLS handshake test. Source: app/config.py — `Settings`.

Fix/result: Not implemented beyond the new safe-default gate. Require partner-approved sources, data-sharing policy, test systems and intended-deployment verification.

QA-F04 — P2 — PARTIAL

DEAD delivery and unknown-profile queues lack controlled recovery actions.

Evidence: HR can inspect persisted failures/mappings, but there is no authorized replay/remapping endpoint or operator procedure. Source: app/routers/admin.py.

Fix/result: Not implemented. Add audited HR-only replay preserving immutable idempotency keys, after reconciling existing gateway state; require explicit mapping decisions.

QA-F05 — P2 — FAIL

Identical advance submissions can create duplicate business requests.

Evidence: Disposable HTTP probe returns201/201 and 2 distinct records. Vacation overlap and document idempotency do not protect initial advance submission. Source: app/services/request_service.py — `create_request`.

Fix/result: Not implemented. Define a client-generated submission key, employee-scoped unique constraint, immutable-body check and UI key reuse on network retry; test serial/concurrent repeats.

QA-F06 — P2 — FAIL

Keyboard focus leaves an open modal.

Evidence: Desktop UI: open request dialog and press Tab 12 times; footer Home gains focus while the dialog remains open. Source: static/js/app.js — `setupModals`.

Fix/result: Not implemented. Trap focus in the top dialog, make the background inert, restore opener focus, and verify keyboard/Escape behavior.

QA-F07 — P2 — FAIL

Live smoke script is not safely repeatable.

Evidence: First 10-step smoke exits0, immediate rerun exits1 on fixed2026-11-23..25 dates with 409 overlap. Source: smoke_test.py — `main`.

Fix/result: Not implemented. Use a fresh disposable fixture per smoke run or explicitly reuse the same known synthetic scenario; do not silently delete user requests.

QA-F08 — P2 — PARTIAL

Database domain constraints do not fully protect direct writes.

Evidence: Disposable SQLite accepts negative leave balance, reversed dates and unknown request status; foreign keys and uniqueness do reject violations. Source: app/models/request.py — `Request`.

Fix/result: Not implemented. Design additive CHECK/enum/decimal hardening with preflight detection and migration tests; do not silently repair existing rows.

QA-F09 — P2 — PARTIAL

Broader AI policy conflicts and hostile-source evaluation are incomplete.

Evidence: Numeric guard catches literal statements differing only in numeric values. Reworded or nonnumeric contradictions are not reliably detected; malicious approved-source suite was not executed. Source: app/services/knowledge_conflicts.py — `has_conflicting_numeric_claims`.

Fix/result: Not implemented beyond F01. Establish HR-controlled policy precedence/version/effective-date metadata and adversarial acceptance fixtures; continue safe fallback under ambiguity.

QA-F10 — P2 — PARTIAL

Operational coverage remains local and incomplete.

Evidence: No CI configuration; no independent app-worker process concurrency run, distributed login limiter, mobile run, production capacity/backup/restore test, or versioned gateway schema upgrades. Source: app/gateway/runtime.py — `initialize`.

Fix/result: Not implemented. Add CI checks and operational tests/runbooks before production

deployment; keep initial create_all limitation explicit.

QA-F11 — P3 — PARTIAL

Some UI/localization/provenance behavior remains incomplete. Evidence: Browser reload clears the in-memory token/navigation; new integration notices retain English text in RU UI; courses rely on global synthetic labeling. Source: static/js/app.js — `showLogin`.

Fix/result: Not implemented. Agree intended session recovery behavior, translate integration messages and add explicit sample provenance where content is shown.

Implemented source changes are limited to the numeric-conflict helper, the AI call orchestration/provider boundary and default-off approval configuration, corresponding environment/Compose wiring and regression tests. README/architecture documentation now describe the actual data flow; the earlier gateway report now correctly records its committed/pushed state. No schema migration or live connector was added.

After F01:75 focused tests passed. After F02:79 AI/knowledge/QA tests and the explicit transport-log-redaction check passed; both final full400-test suites passed. No implementation changes were made for the remaining P2/P3 findings; acceptance criteria are in FIX_PLAN.md.

Boundaries and next actions

This review does not establish production readiness, vendor/model availability, official policy approval, legal/data-residency compliance, production mTLS, HA/capacity, backup restoration, mobile accessibility or exhaustive button coverage. Internal HTTP is confined to the local MOCK setup. Existing data and secret environment files were preserved.

1. Approve intended deployment/source-sharing policy and partner test contracts; keep MOCK/offline defaults until independently verified.
2. Add audited recovery/remapping and initial submission idempotency; then exercise separate worker processes and migration/restore operations.
3. Fix dialog focus/repeatable smoke, add CI and run the unverified browser/mobile/production-like checks.

Preview of the inspected source-backed answer: ![Synthetic AI source citation](docs/evidence/qa-2026-10-17/ai-source.png)
\# JanaМен QA checklist — 17 October 2026\
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Final-state assessment of all **162** checkbox/numbered acceptance items from the supplied checklist. PASS: 140 / FAIL: 2 / PARTIAL: 17 / NOT VERIFIED: 2 / BLOCKED: 0 / NOT APPLICABLE: 1.\
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PASS is scoped to the evidence stated. Real vendor connections, production infrastructure and live Gemini are not established by MOCK tests. All 16 required acceptance scenarios are covered by executed tests; broader operational gaps remain.\
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\## PHASE 1: REPOSITORY DISCOVERY\
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\- **QA001 — PASS** — Repository structure and entry points\
  Evidence: Inspected [app/main.py]\(app/main.py),

[app/gateway/main.py]\(app/gateway/main.py), [static/index.html]\(static/index.html).\
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\- **QA002 — PASS** — Frontend framework and build commands\
  Evidence: Inspected [static/js/app.js]\(static/js/app.js),
[static/css/style.css]\(static/css/style.css), [static/js/i18n.js]\(static/js/i18n.js). Plain
HTML/CSS/JavaScript; no package.json, frontend build or bundler.\
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\- **QA003 — PASS** — Backend framework and startup commands\
  Evidence: Inspected [app/main.py]\(app/main.py), [Dockerfile]\(Dockerfile).\
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\- **QA004 — PASS** — Database engine, models, and migrations\
  Evidence: Inspected [app/database.py]\(app/database.py),
[app/models/request.py]\(app/models/request.py),
[migrations/versions/0006_gateway_runtime.py]\(migrations/versions/0006_gateway_runtime.py).\
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\- **QA005 — PASS** — Authentication and authorization implementation\
  Evidence: Inspected [app/services/auth_service.py]\(app/services/auth_service.py),
[app/routers/auth.py]\(app/routers/auth.py).\
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\- **QA006 — PASS** — API routes and validation\
  Evidence: Inspected [app/routers/requests.py]\(app/routers/requests.py),
[app/schemas/request.py]\(app/schemas/request.py), [docs/openapi.json]\(docs/openapi.json).\
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\- **QA007 — PASS** — Background workers and scheduled tasks\
  Evidence: Inspected [app/integrations/outbox.py]\(app/integrations/outbox.py),
[app/gateway/runtime.py]\(app/gateway/runtime.py).\
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\- **QA008 — PASS** — Integration adapters and gateway\
  Evidence: Inspected [app/gateway/connectors.py]\(app/gateway/connectors.py),
[app/integrations/profile_sync.py]\(app/integrations/profile_sync.py).\
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\- **QA009 — PASS** — AI assistant implementation and knowledge sources\
  Evidence: Inspected [app/services/ai_service.py]\(app/services/ai_service.py),
[knowledge/README.md]\(knowledge/README.md).\
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\- **QA010 — PASS** — Unit, integration, and end-to-end tests\
  Evidence: Inspected [tests/test_e2e_journey.py]\(tests/test_e2e_journey.py),
[tests/test_gateway_service.py]\(tests/test_gateway_service.py),
[tests/frontend_localization.test.cjs]\(tests/frontend_localization.test.cjs).\
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\- **QA011 — PASS** — Dockerfiles and Docker Compose\
  Evidence: Inspected [Dockerfile]\(Dockerfile), [docker-compose.yml]\(docker-compose.yml).\
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\- **QA012 — PARTIAL** — CI configuration and lint/type-check commands\
  Evidence: [pyproject.toml]\(pyproject.toml) defines Ruff/mypy/pytest; no tracked CI
workflow or .github directory was found.\
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\- **QA013 — PASS** — README and architecture documentation\
  Evidence: Inspected [README.md]\(README.md),
[ARCHITECTURE.md]\(ARCHITECTURE.md).\
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\- **QA014 — PASS** — Environment variable examples\
  Evidence: Inspected [.env.example]\(.env.example), [app/config.py]\(app/config.py).\
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\- **QA015 — PASS** — Existing TODOs, placeholders, disconnected services, and mock data\
  Evidence: Inspected
[app/integrations/documentolog.py]\(app/integrations/documentolog.py),
[app/integrations/one_c_zup.py]\(app/integrations/one_c_zup.py),
[app/gateway/mock_employees.json]\(app/gateway/mock_employees.json). Legacy adapters are disconnected simulations; current gateway fixtures are explicitly MOCK. Graph text search found no TODO/FIXME/NotImplementedError matches; absence of markers does not prove completeness.\
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\## PHASE 2: END-TO-END PRODUCT REQUIREMENTS\
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\### Employee workspace\
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\- **QA016 — PASS** — Employee can view their own profile.\
  Evidence: [test_get_my_profile]\(tests/test_auth.py#L42); [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
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\- **QA017 — PASS** — Official HR profile fields are read-only for employees.\
  Evidence: [test_local_provisioning_creates_readonly_profile]\(tests/test_reliability.py#L555); [test_matrix_covers_every_openapi_operation]\(tests/test_api_matrix.py#L121); no employee profile mutation route exists.\
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\- **QA018 — PASS** — Employee can submit a supported HR request.\
  Evidence: [test_create_vacation_request_success]\(tests/test_requests.py#L11); [test_create_advance_payment_request]\(tests/test_requests.py#L52); [test_full_hackathon_demo_journey]\(tests/test_e2e_journey.py#L1); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44)\
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\- **QA019 — PASS** — Input validation rejects invalid or incomplete requests.\
  Evidence:
[test_body_contracts_reject_missing_malformed_and_extra_input]\(tests/test_api_matrix.py#L227); [test_request_invalid_inputs]\(tests/test_reliability.py#L108)\
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\- **QA020 — PASS** — Request is persisted in the database.\
  Evidence: [test_create_vacation_request_success]\(tests/test_requests.py#L11); [test_create_advance_payment_request]\(tests/test_requests.py#L52); [test_full_hackathon_demo_journey]\(tests/test_e2e_journey.py#L1); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44)\
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\- **QA021 — PASS** — Employee can see their own request history.\
  Evidence: [test_create_vacation_request_success]\(tests/test_requests.py#L11); [test_create_advance_payment_request]\(tests/test_requests.py#L52); [test_full_hackathon_demo_journey]\(tests/test_e2e_journey.py#L1); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44)\
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\- **QA022 — PASS** — Request status and history remain consistent after reload.\
  Evidence: [test_create_vacation_request_success]\(tests/test_requests.py#L11); [test_create_advance_payment_request]\(tests/test_requests.py#L52); [test_full_hackathon_demo_journey]\(tests/test_e2e_journey.py#L1);

[test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44)\
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\- **QA023 — PASS** — Employee cannot view another employee's private requests.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132)\
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\- **QA024 — PARTIAL** — Loading, empty, error, and success states are handled.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json); [static/js/app.js]\(static/js/app.js) has errors, empty messages and disabled controls; browser network-error/loading permutations were not exhaustively tested.\
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\- **QA025 — PASS** — Displayed leave balances show their source and synchronization freshness.\
  Evidence: [test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
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\### Manager approvals\
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\- **QA026 — PASS** — Manager can see only requests within their authorized scope.\   Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132); [test_manager_review_direct_report_approval]\(tests/test_requests.py#L66); [test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184); [test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44); [app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA027 — PASS** — Manager can approve or reject eligible requests.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132); [test_manager_review_direct_report_approval]\(tests/test_requests.py#L66); [test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184); [test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44); [app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA028 — PASS** — Unauthorized roles cannot approve requests.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132); [test_manager_review_direct_report_approval]\(tests/test_requests.py#L66); [test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184); [test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44); [app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA029 — PASS** — Approval/rejection creates a history record.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132); [test_manager_review_direct_report_approval]\(tests/test_requests.py#L66); [test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184); [test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207); [test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44); [app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA030 — PASS** — Audit events capture important state transitions.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_review_direct_report_approval]\(tests/test_requests.py#L66);
[test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184);
[test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207);
[test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44);
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA031 — PASS** — Notifications are created once per relevant event.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_review_direct_report_approval]\(tests/test_requests.py#L66);
[test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184);
[test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207);
[test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44);
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA032 — PASS** — Concurrent actions cannot create contradictory final states.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_review_direct_report_approval]\(tests/test_requests.py#L66);
[test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184);
[test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207);
[test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44);
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385)\
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\- **QA033 — PASS** — A repeated approval action is handled idempotently.\   Evidence:
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_review_direct_report_approval]\(tests/test_requests.py#L66);
[test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184);
[test_concurrent_approvals_cannot_overdraw]\(tests/test_reliability.py#L207);
[test_history_shows_decisions_with_actor_and_comment]\(tests/test_seed_scale.py#L44);
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385) A repeated decision returns 409 with no extra business effects; it does not return a second success.\
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\### Jana Academy\
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\- **QA034 — PASS** — Course catalogue loads from the actual configured source.\
  Evidence: [test_list_courses]\(tests/test_academy.py#L1);
[test_course_catalog_uses_bounded_queries]\(tests/test_backend_hardening.py#L334); server database catalogue, not an external LMS.\
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\- **QA035 — PASS** — Course enrollment or assignment works as designed.\
  Evidence:
[test_explicit_enrollment_progress_and_persistence]\(tests/test_reliability.py#L283);
[test_concurrent_enrollment_is_idempotent]\(tests/test_reliability.py#L495)\
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\- **QA036 — PASS** — Lesson progress is persisted.\
  Evidence:
[test_explicit_enrollment_progress_and_persistence]\(tests/test_reliability.py#L283);
[test_concurrent_enrollment_is_idempotent]\(tests/test_reliability.py#L495)\
\
\- **QA037 — PASS** — Progress survives reload and re-login.\

  Evidence:
[test_explicit_enrollment_progress_and_persistence]\(tests/test_reliability.py#L283); [relogin-persistence.json]\(docs/evidence/qa-2026-10-17/relogin-persistence.json) records actual HTTP re-login and state after the QA web process rebuild.\
\
\- **QA038 — PASS** — Completion state is correct.\
  Evidence: [test_complete_lesson_quiz_and_finish_course]\(tests/test_academy.py#L31); [test_course_completion_notifies_once]\(tests/test_reliability.py#L326)\
\
\- **QA039 — PASS** — Course access and progress are scoped to the authenticated employee.\
  Evidence:
[app/services/academy_service.py:get_enrollment]\(app/services/academy_service.py#L24); [test_each_operation_has_a_working_authorized_request]\(tests/test_api_matrix.py#L140); [test_explicit_enrollment_progress_and_persistence]\(tests/test_reliability.py#L283)\
\
\- **QA040 — PARTIAL** — Sample course content is identified as sample content.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json); global synthetic-demo footer exists, but course cards have no explicit per-course sample provenance field.\
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\### AI Jana\
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\- **QA041 — PASS** — Assistant retrieves only approved, authorized knowledge sources.\
  Evidence:
[test_document_approval_lifecycle_and_source_scope]\(tests/test_reliability.py#L381); [test_import_review_and_ml_citations_preserve_authorization]\(tests/test_knowledge_files_and_ml.py#L133)\
\
\- **QA042 — PASS** — Answers include verifiable source references.\
  Evidence:
[test_citation_rejects_changed_source_after_reapproval]\(tests/test_backend_hardening.py#L145); [ai-source.png]\(docs/evidence/qa-2026-10-17/ai-source.png)\
\
\- **QA043 — PASS** — Unsupported questions receive a safe fallback.\
  Evidence: [test_ai_safe_fallback_unsupported_question]\(tests/test_ai.py#L28); [test_ai_safe_fallback_russian]\(tests/test_ai.py#L93); [test_ai_safe_fallback_kazakh]\(tests/test_ai.py#L164)\
\
\- **QA044 — PARTIAL** — Retrieved content cannot override system security rules.\
  Evidence:
[test_file_cannot_set_approval_or_disable_source_rules]\(tests/test_knowledge_files_and_ml.py#L56); [test_valid_quote_does_not_authorize_fabricated_claim]\(tests/test_merge_regressions.py#L27); exact quotation/role filters are enforced. A comprehensive malicious-approved-source injection evaluation was not executed.\
\
\- **QA045 — PARTIAL** — Assistant does not invent company policies or HR contacts.\   Evidence:
[test_conflicting_approved_numbers_fall_back_before_provider]\(tests/test_qa_acceptance.py#L44); [test_unverified_model_answer_falls_back_to_source_text]\(tests/test_ai_gemini.py#L73); reproduced numeric disagreement now falls back. General semantic contradictions and employer approval of configured HR contact remain unverified.\

\
\- **QA046 — PASS** — Private employee records are not leaked to other users.\
  Evidence: [test_ai_privacy_guardrail_blocked]\(tests/test_ai.py#L41); [test_private_questions_never_reach_the_model]\(tests/test_ai_gemini.py#L99); no employee-record retrieval tool is exposed to the provider.\
\
\- **QA047 — PARTIAL** — Sensitive personal data is not sent to an AI provider unless explicitly approved and protected by the system's data policy.\
  Evidence: [test_employee_free_text_is_absent_from_provider_http_body]\(tests/test_ai_gemini.py#L146); [test_provider_requires_explicit_approval_at_http_boundary]\(tests/test_ai_gemini.py#L126); raw questions stay local and approval defaults off. Actual organizational approval, provider retention/residency and PII in approved source text are NOT VERIFIED.\
\
\- **QA048 — PASS** — Provider failures, empty retrieval results, and timeouts are handled.\
  Evidence: [test_model_outage_keeps_extractive_answers]\(tests/test_ai_gemini.py#L93); [test_out_of_scope_question_stays_a_safe_fallback]\(tests/test_ai_gemini.py#L108); [test_gemini_error_logs_are_redacted]\(tests/test_merge_regressions.py#L57); provider transport is intercepted; live Gemini remains NOT VERIFIED.\
\
\- **QA049 — PASS** — The interface accurately identifies demo or production mode.\
  Evidence: [test_production_settings_and_synthetic_boundaries]\(tests/test_reliability.py#L459); [test_published_demo_password_is_denied_in_production]\(tests/test_reliability.py#L602); [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
\
\- **QA050 — PARTIAL** — Language handling matches the implemented RU/KZ/English requirements.\
  Evidence: [test_ai_grounded_answer_russian]\(tests/test_ai.py#L72); [test_ai_grounded_answer_kazakh]\(tests/test_ai.py#L135); [test_ai_grounded_answer_vacation]\(tests/test_ai.py#L4); [test_language_negotiation]\(tests/test_localization.py#L179); RU/KK frontend and EN/RU/KK backend tested. New integration notifications and some labels remain English in the RU UI.\
\
\### Notifications\
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\- **QA051 — PASS** — Correct user receives each notification.\
  Evidence: [test_notification_ownership_and_unread]\(tests/test_reliability.py#L265); [test_course_completion_notifies_once]\(tests/test_reliability.py#L326); [test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79)\
\
\- **QA052 — PASS** — Unauthorized users cannot read other users' notifications.\
  Evidence: [test_notification_ownership_and_unread]\(tests/test_reliability.py#L265); [test_course_completion_notifies_once]\(tests/test_reliability.py#L326); [test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79)\
\
\- **QA053 — PASS** — Read/unread status persists.\
  Evidence: [test_notification_ownership_and_unread]\(tests/test_reliability.py#L265); [test_course_completion_notifies_once]\(tests/test_reliability.py#L326); [test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79)\
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\- **QA054 — PASS** — Duplicate events do not generate duplicate notifications.\
  Evidence: [test_notification_ownership_and_unread]\(tests/test_reliability.py#L265);
[test_course_completion_notifies_once]\(tests/test_reliability.py#L326);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79)\
\
\- **QA055 — PASS** — Failed notification delivery is observable and recoverable.\
Evidence: [test_failed_notification_rolls_back_entire_approval]\(tests/test_reliability.py#L245);
[test_failed_approval_rolls_back_outbox_and_all_business_effects]\(tests/test_qa_acceptance.py#L89)
; delivery here means persistent in-app notification rows. A failed write aborts the business
transaction and returns an error, allowing the action to be retried; email/push delivery is not
implemented.\
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\## PHASE 3: AUTHENTICATION AND AUTHORIZATION\
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\- **QA056 — PASS** — Authentication is required for protected API routes.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA057 — PASS** — Employee, manager, and HR permissions are enforced by the backend.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA058 — PASS** — Role changes cannot be achieved by editing client-side state.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA059 — PASS** — Object-level access checks prevent IDOR vulnerabilities.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA060 — PASS** — Employees cannot access manager or HR endpoints without permission.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);

[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA061 — PASS** — Manager scope is enforced server-side.\
  Evidence:
[test_every_protected_operation_rejects_anonymous_access]\(tests/test_api_matrix.py#L174);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183);
[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA062 — PASS** — Session/token expiration and invalid credentials are handled.\
  Evidence:
[test_all_protected_operations_reject_invalid_sessions]\(tests/test_api_matrix.py#L196);
[test_logout_revokes_current_session_only]\(tests/test_reliability.py#L27);
[test_zero_lifetime_token_expires_instead_of_using_default]\(tests/test_backend_hardening.py#L235)
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\- **QA063 — PASS** — Passwords, if supported, are stored using an appropriate password-hashing scheme.\   Evidence:
[app/services/auth_service.py:hash_password]\(app/services/auth_service.py#L73);
[test_legacy_password_is_upgraded_after_login]\(tests/test_reliability.py#L535)\
\
\- **QA064 — PASS** — Secrets are not committed to source control.\
  Evidence: [secret-scan.json]\(docs/evidence/qa-2026-10-17/secret-scan.json); scoped current-file and reachable-history scan; private environment files remain ignored. A scan is not proof that every possible secret format is absent.\
\
\- **QA065 — PARTIAL** — CORS, CSRF, cookie, and rate-limit settings are reviewed where applicable.\
  Evidence: [app/main.py]\(app/main.py);
[test_login_throttle_and_redacted_errors]\(tests/test_reliability.py#L74);
[test_production_settings_and_synthetic_boundaries]\(tests/test_reliability.py#L459); explicit CORS origins and bearer Authorization reviewed. No auth cookies (cookie CSRF not applicable); login throttle is per-process, not distributed.\
\
\- **QA066 — PASS** — Authentication failures do not reveal sensitive information.\
  Evidence: [test_login_invalid_password]\(tests/test_auth.py#L13);
[test_login_nonexistent_user]\(tests/test_auth.py#L21);
[test_login_throttle_and_redacted_errors]\(tests/test_reliability.py#L74)\
\
\- **QA067 — PASS** — Logs avoid secrets and unnecessary personal data.\
  Evidence:
[test_ai_audit_is_persisted_without_raw_questions]\(tests/test_reliability.py#L362);
[test_gemini_error_logs_are_redacted]\(tests/test_merge_regressions.py#L57);
[test_unexpected_error_is_redacted_and_not_cached]\(tests/test_backend_hardening.py#L192); [log-hygiene.json]\(docs/evidence/qa-2026-10-17/log-hygiene.json)\
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\## PHASE 4: DOCUMENTOLOG ASYNCHRONOUS INTEGRATION\
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\### Transaction and outbox\

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\- **QA068 — PASS** — Approval and outbox insertion occur in the same database transaction.\
  Evidence:
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385);
[test_failed_approval_rolls_back_outbox_and_all_business_effects]\(tests/test_qa_acceptance.py#L89)
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\- **QA069 — PASS** — If the transaction fails, neither partial approval nor orphan outbox event remains.\
  Evidence:
[test_failed_approval_rolls_back_outbox_and_all_business_effects]\(tests/test_qa_acceptance.py#L89) injects failure after enqueue+flush, during notification and during audit; asserts Pending, unchanged balance/history/notification counts, no outbox row.\
\
\- **QA070 — PASS** — External network calls do not occur inside the approval transaction.\
  Evidence:
[app/services/request_service.py:review_request]\(app/services/request_service.py#L385) only enqueues a snapshot; network delivery is in
[app/integrations/outbox.py:dispatch_one]\(app/integrations/outbox.py#L242).\
\
\- **QA071 — PASS** — Approval remains successful if Documentolog or the gateway is unavailable.\
  Evidence:
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109)
; [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA072 — PASS** — Worker safely claims pending rows.\
  Evidence: [app/integrations/outbox.py:dispatch_one]\(app/integrations/outbox.py#L242) uses PostgreSQL FOR UPDATE SKIP LOCKED;
[app/gateway/runtime.py:process_one]\(app/gateway/runtime.py#L124) uses conditional durable leases. Intended application Compose database is PostgreSQL.\
\
\- **QA073 — PARTIAL** — Concurrent workers cannot process the same event unsafely.\
  Evidence:
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124) verifies gateway job/document concurrency; PostgreSQL row-claim SQL inspected. Simultaneous independent application-worker processes were not exercised; SQLite application workers do not gain PostgreSQL row locks.\
\
\- **QA074 — PASS** — Delivery attempts and error reasons are recorded.\   Evidence:
[app/integrations/outbox.py:_fail]\(app/integrations/outbox.py#L218);
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109)
; [test_malformed_ack_is_not_success]\(tests/test_integration_gateway.py#L135)\
\
\- **QA075 — PASS** — Retry behavior uses bounded exponential backoff with jitter.\
  Evidence: [app/integrations/outbox.py:_fail]\(app/integrations/outbox.py#L218) and [app/gateway/runtime.py:process_one]\(app/gateway/runtime.py#L124) use a bounded increasing schedule with uniform 0.8–1.2 jitter; retry exhaustion executed in tests. Day-long real elapsed schedules were not waited through.\
\
\- **QA076 — PASS** — Maximum retries transition to a visible DEAD state.\

  Evidence:
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109)
;
[test_mock_connector_temporary_failure_reaches_dead_without_document]\(tests/test_gateway_servi
ce.py#L207)\
\
\- **QA077 — PARTIAL** — Dead events can be inspected and safely retried by authorized staff.\
  Evidence:
[test_validation_failure_not_retried_and_hr_queue_shows_it]\(tests/test_integration_gateway.py#L14
5); HR can inspect FAILED/DEAD rows, but there is no authorized replay endpoint/button/runbook.
Do not reset rows blindly.\
\
\- **QA078 — PASS** — Worker restart does not silently lose jobs.\
  Evidence:
[test_gateway_restart_recovers_jobs_and_nonces]\(tests/test_gateway_service.py#L140);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[test_reconciliation_recovers_lost_callback]\(tests/test_integration_gateway.py#L159); actual Docker
rebuild preserved registered records; no production backup/restore claim.\
\
\### Idempotency\
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\- **QA079 — PASS** — Each logical event has a stable idempotency key.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
\
\- **QA080 — PASS** — The same key cannot create multiple external documents.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
\
\- **QA081 — PASS** — Database uniqueness constraints enforce deduplication.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
\
\- **QA082 — PASS** — Retries reuse the original key.\   Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);

[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
\
\- **QA083 — PASS** — Duplicate callbacks do not repeat state changes or notifications.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
\
\- **QA084 — PASS** — Callback processing is safe if the same logical result arrives again.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[test_callback_retry_is_persistent_and_never_registers_twice]\(tests/test_gateway_service.py#L166);
[app/models/integration.py]\(app/models/integration.py) and
[app/gateway/models.py]\(app/gateway/models.py) enforce unique keys.\
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\### Request signing and callbacks\
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\- **QA085 — PASS** — HMAC-SHA256 signing matches the documented canonical message format.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA086 — PASS** — Method, path, timestamp, nonce, and body hash are included consistently.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA087 — PASS** — Invalid signatures are rejected.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);

[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA088 — PASS** — Expired or excessively skewed timestamps are rejected.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA089 — PASS** — Reused nonces are rejected.\   Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA090 — PASS** — Unknown key IDs are rejected.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py#L60);
[test_query_string_is_signed_and_mock_snapshots_are_canonical]\(tests/test_gateway_service.py#L184);
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174); [app/integrations/signing.py]\(app/integrations/signing.py)\
\
\- **QA091 — PASS** — Signature comparison uses a constant-time comparison.\
  Evidence: [app/integrations/signing.py]\(app/integrations/signing.py) uses hmac.compare_digest; signature rejection/tampering tests execute that verifier.\
\
\- **QA092 — PASS** — Callback authorization is independent of frontend roles.\
  Evidence:
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[app/routers/internal.py:documentolog_status]\(app/routers/internal.py#L47) and
[app/integrations/outbox.py:apply_result]\(app/integrations/outbox.py#L131)\
\
\- **QA093 — PASS** — Callback payloads are validated before state changes.\
  Evidence:
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[app/routers/internal.py:documentolog_status]\(app/routers/internal.py#L47) and
[app/integrations/outbox.py:apply_result]\(app/integrations/outbox.py#L131)\
\
\- **QA094 — PASS** — Callback processing is idempotent and transactional.\
  Evidence:

[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174);
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79);
[app/routers/internal.py:documentolog_status]\(app/routers/internal.py#L47) and
[app/integrations/outbox.py:apply_result]\(app/integrations/outbox.py#L131)\
\
\- **QA095 — PASS** — Secrets are loaded from protected configuration.\
  Evidence: [app/config.py]\(app/config.py);
[app/gateway/configure_mock.py]\(app/gateway/configure_mock.py);
[test_invalid_key_errors_hide_input_values]\(tests/test_gateway_service.py#L225);
[test_short_integration_keys_are_refused]\(tests/test_integration_gateway.py#L66); generated env is outside the repo, mode0600.\
\
\- **QA096 — PARTIAL** — HTTPS/mTLS requirements are documented and enforced in the intended deployment.\
  Evidence: [docs/INTEGRATIONS.md]\(docs/INTEGRATIONS.md);
[app/integrations/outbox.py:_tls]\(app/integrations/outbox.py#L86) and production config validation. Local MOCK uses internal HTTP+HMAC; inbound mTLS proxy, certificates and production handshake are NOT VERIFIED.\
\
\- **QA097 — PASS** — No custom cryptographic algorithm has been invented.\
  Evidence: [app/integrations/signing.py]\(app/integrations/signing.py) uses standard HMAC-SHA256/base64/SHA256, with method/path-query/time/nonce/body digest binding; no custom cipher.\
\
\### Mock Documentolog behavior\
\
\- **QA098 — PASS** — Mock connector is explicitly labelled MOCK.\
  Evidence: [app/gateway/connectors.py]\(app/gateway/connectors.py); [hr-card.png]\(docs/evidence/qa-2026-10-17/hr-card.png)\
\
\- **QA099 — PASS** — Successful mock registration returns a deterministic external reference.\
  Evidence: [app/gateway/connectors.py:register]\(app/gateway/connectors.py#L26) derives MOCK-DOC from the persisted job ID; repeated key/restart returns the same stored reference. Recreating an empty store is not the same persisted logical job.\
\
\- **QA100 — PASS** — The gateway acknowledges accepted work asynchronously.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[app/gateway/main.py:documents]\(app/gateway/main.py#L95) returns 202; registration runs in the gateway worker.\
\
\- **QA101 — PASS** — The mock can simulate temporary failure and validation rejection.\
  Evidence:
[test_mock_connector_temporary_failure_reaches_dead_without_document]\(tests/test_gateway_service.py#L207);
[test_validation_failure_not_retried_and_hr_queue_shows_it]\(tests/test_integration_gateway.py#L145)\
\
\- **QA102 — PASS** — A successful callback updates the JanaМен request integration state.\
  Evidence:

[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA103 — PASS** — Employee approval status remains Approved when document registration fails.\
  Evidence:
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109);
[test_validation_failure_not_retried_and_hr_queue_shows_it]\(tests/test_integration_gateway.py#L145)\
\
\- **QA104 — PASS** — UI distinguishes NOT_REQUIRED, PENDING, SENT, REGISTERED, FAILED, and DEAD as applicable.\
  Evidence: [app/schemas/request.py]\(app/schemas/request.py);
[static/js/app.js]\(static/js/app.js) renders sync_status/external_ref independently of business status;
[test_without_gateway_approval_queues_nothing]\(tests/test_integration_gateway.py#L71);
[test_validation_failure_not_retried_and_hr_queue_shows_it]\(tests/test_integration_gateway.py#L145);
[test_admin_card_uses_actual_counts_mode_and_last_success]\(tests/test_integration_gateway.py#L284). Browser observed REGISTERED/MOCK and pending/failed counters; not every status was visually injected.\
\
\- **QA105 — PASS** — Real Documentolog connectivity is not claimed without a verified vendor connection.\
  Evidence: [docs/INTEGRATIONS.md]\(docs/INTEGRATIONS.md) and [hr-card.png]\(docs/evidence/qa-2026-10-17/hr-card.png) explicitly state MOCK, with no live vendor registration/signature claim.\
\
\## PHASE 5: 1С:ЗУП INTEGRATION\
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\- **QA106 — PASS** — Employee records are matched using a stable employee code, not name alone.\
  Evidence:
[test_profile_sync_code_matching_and_provisional_balance]\(tests/test_integration_gateway.py#L211);
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json);
[browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
\
\- **QA107 — PASS** — Synced records identify their source and synchronization timestamp.\
  Evidence:
[test_profile_sync_code_matching_and_provisional_balance]\(tests/test_integration_gateway.py#L211);
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json);
[browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
\
\- **QA108 — PASS** — Employee snapshots remain available during temporary gateway outages.\
  Evidence:
[test_profile_sync_code_matching_and_provisional_balance]\(tests/test_integration_gateway.py#L21

1);
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json);
[browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
\
\- **QA109 — PASS** — Stale data is visibly marked.\   Evidence:
[test_profile_sync_code_matching_and_provisional_balance]\(tests/test_integration_gateway.py#L211);
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json);
[browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json)\
\
\- **QA110 — PARTIAL** — Unmatched employee records enter a controlled HR review process.\
  Evidence:
[test_unmatched_employee_and_manager_go_to_review]\(tests/test_integration_gateway.py#L235);
safe persisted HR review list exists, but resolution/remapping workflow is not implemented.\
\
\- **QA111 — PASS** — Read-only 1C access is used where specified.\
  Evidence: [app/gateway/connectors.py]\(app/gateway/connectors.py) reads synthetic snapshot fixtures only. No write-back method or real vendor API is implemented.\
\
\- **QA112 — PASS** — Leave balance source and calculation ownership are documented.\
  Evidence: [docs/INTEGRATIONS.md]\(docs/INTEGRATIONS.md) documents snapshot source balance minus frozen unconfirmed local debits and the explicit source-confirmation extension;
[test_snapshot_failure_does_not_confirm_debits_or_overwrite_balance]\(tests/test_integration_gateway.py#L303)\
\
\- **QA113 — PASS** — Failed synchronization can be retried safely.\
  Evidence:
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); HR-only, rate-limited refresh and scheduled retry keep last-known snapshot.\
\
\- **QA114 — PASS** — No duplicate employees are created by repeated sync.\
  Evidence:
[test_profile_sync_code_matching_and_provisional_balance]\(tests/test_integration_gateway.py#L211);
[test_unmatched_employee_and_manager_go_to_review]\(tests/test_integration_gateway.py#L235);
existing employee codes only; no account auto-creation.\
\
\- **QA115 — PASS** — MOCK and CONNECTED modes are visibly distinguished.\
  Evidence: [hr-card.png]\(docs/evidence/qa-2026-10-17/hr-card.png);
[test_admin_card_uses_actual_counts_mode_and_last_success]\(tests/test_integration_gateway.py#L284); only MOCK or UNCONNECTED is reported, never an unverified CONNECTED state.\
\
\- **QA116 — PASS** — Live 1C connectivity is not claimed without a successful integration test.\
  Evidence: [docs/INTEGRATIONS.md]\(docs/INTEGRATIONS.md); no live 1C claim; real connectivity remains NOT VERIFIED.\
\
\## PHASE 6: DATABASE AND API QUALITY\
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\- **QA117 — PASS** — Migrations apply cleanly in a disposable test database.\
  Evidence: [tests/test_migrations.py]\(tests/test_migrations.py) contains 6 migration cases; the full runs also migrate fresh SQLite databases and fresh PostgreSQL schemas for application tests.\
\
\- **QA118 — PARTIAL** — Foreign keys and uniqueness constraints protect data integrity.\
  Evidence: [extra-probes.json]\(docs/evidence/qa-2026-10-17/extra-probes.json) rejects FK and duplicate employee code violations, but direct SQLite writes accept negative balance, reversed dates and unknown request status. API validation protects normal calls; DB CHECK hardening remains.\
\
\- **QA119 — PARTIAL** — Important queries have appropriate indexes.\
  Evidence: [app/models/request.py]\(app/models/request.py), [app/models/profile.py]\(app/models/profile.py), [app/models/integration.py]\(app/models/integration.py) include ownership/status/code/due indexes; [test_course_catalog_uses_bounded_queries]\(tests/test_backend_hardening.py#L334). Production query plans/capacity were not benchmarked in this QA run.\
\
\- **QA120 — PASS** — API inputs are validated.\
  Evidence: [test_body_contracts_reject_missing_malformed_and_extra_input]\(tests/test_api_matrix.py#L227); [test_all_collections_enforce_pagination_bounds]\(tests/test_backend_hardening.py#L297)\
\
\- **QA121 — PASS** — API errors are consistent and do not leak internals.\   Evidence: [test_unexpected_error_is_redacted_and_not_cached]\(tests/test_backend_hardening.py#L192); [test_openapi_documents_response_and_error_shapes]\(tests/test_reliability.py#L485); [test_canonical_validation_does_not_echo_payload]\(tests/test_gateway_service.py#L196)\
\
\- **QA122 — PASS** — Pagination and filtering behave correctly where needed.\
  Evidence: [test_collections_have_stable_pagination]\(tests/test_backend_hardening.py#L280); [test_all_collections_enforce_pagination_bounds]\(tests/test_backend_hardening.py#L297); [test_each_operation_has_a_working_authorized_request]\(tests/test_api_matrix.py#L140)\
\
\- **QA123 — PASS** — State transitions reject invalid operations.\
  Evidence: [test_transition_and_action_validation]\(tests/test_reliability.py#L156); [test_concurrent_duplicate_approval]\(tests/test_reliability.py#L184)\
\
\- **QA124 — PASS** — Database transactions cover logically atomic updates.\
  Evidence: [test_failed_approval_rolls_back_outbox_and_all_business_effects]\(tests/test_qa_acceptance.py#L89); [test_failed_knowledge_edit_rolls_back_revision]\(tests/test_backend_hardening.py#L245)\
\
\- **QA125 — PASS** — Timezones and dates are handled consistently.\
  Evidence: [test_timestamps_are_unambiguous_utc]\(tests/test_reliability.py#L517); [test_language_negotiation]\(tests/test_localization.py#L179); UTC serialization and date/exclusion rules tested. Employer approval of the configured calendar is outside this verification.\
\
\- **QA126 — PASS** — Sensitive fields are excluded from unnecessary API responses.\
  Evidence: [app/schemas/auth.py]\(app/schemas/auth.py), [app/schemas/profile.py]\(app/schemas/profile.py);

[test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57);
[test_employee_free_text_is_absent_from_provider_http_body]\(tests/test_ai_gemini.py#L146);
secrets/password hashes not serialized in user/profile DTOs.\
\
\- **QA127 — FAIL** — Repeated requests do not create unintended duplicate records.\
  Evidence: [extra-probes.json]\(docs/evidence/qa-2026-10-17/extra-probes.json) repeats the identical advance payload twice: HTTP201/201,2 distinct request records. No client request-idempotency contract; vacation overlap and integration keys have separate protection.\
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\## PHASE 7: FRONTEND AND USER EXPERIENCE\
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\- **QA128 — PASS** — Every visible navigation item works.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json) covers6 employee primary pages, manager Team and HR Administration. Duplicate navigation controls share the same go() handler; this does not establish every form/action permutation.\
\
\- **QA129 — NOT VERIFIED** — Every action button has a real, correctly scoped behavior.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json) is a scoped desktop browser check, not exhaustive coverage of every visible button in every role/state.\
\
\- **QA130 — PASS** — Forms validate required fields and date ranges.\
  Evidence: [static/index.html]\(static/index.html) has required fields;
[static/js/app.js]\(static/js/app.js) sets dates and displays API errors;
[test_create_vacation_request_invalid_dates]\(tests/test_requests.py#L27);
[test_dates_must_be_from_today_to_one_year_ahead]\(tests/test_request_rules.py#L23)\
\
\- **QA131 — PARTIAL** — Errors are visible and actionable.\
  Evidence: [static/js/app.js]\(static/js/app.js) displays error/toast text; browser checked sign-in/source/refresh success. Recovery guidance for raw mapping/connector error codes and all network failures is incomplete.\
\
\- **QA132 — PARTIAL** — Loading and empty states exist.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json) observed empty failure queue and disabled refresh; startup briefly displays placeholders. Full loading/failure-state browser coverage is not verified.\
\ \- **QA133 — PARTIAL** — State remains correct after refresh where persistence is expected.\
  Evidence: [relogin-persistence.json]\(docs/evidence/qa-2026-10-17/relogin-persistence.json) verifies server progress survives. Browser reload clears in-memory token and returns to sign-in, losing current navigation; stored business records persist.\
\
\- **QA134 — PASS** — UI status reflects backend truth rather than hard-coded assumptions.\
  Evidence:
[test_admin_card_uses_actual_counts_mode_and_last_success]\(tests/test_integration_gateway.py#L284); [hr-card.png]\(docs/evidence/qa-2026-10-17/hr-card.png)\
\
\- **QA135 — NOT VERIFIED** — Mobile layout is usable.\
  Evidence: No mobile viewport/device browser session was executed in this run; responsive CSS alone is insufficient evidence.\
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\- **QA136 — FAIL** — Keyboard navigation and accessible labels are present.\
  Evidence: [browser-observations.json]\(docs/evidence/qa-2026-10-17/browser-observations.json) reproduces keyboard focus escaping an open request dialog to footer Home after 12 Tab presses. Labels/dialog roles exist; focus trapping/background inertness is absent.\
\
\- **QA137 — PASS** — Mock data is clearly identified.\
  Evidence: [ai-source.png]\(docs/evidence/qa-2026-10-17/ai-source.png), [hr-card.png]\(docs/evidence/qa-2026-10-17/hr-card.png); source disclaimers, MOCK mode, prototype/synthetic footer.\
\
\- **QA138 — PASS** — Role preview controls are never presented as real authentication.\
  Evidence: [static/js/app.js:switchRole]\(static/js/app.js) graph range is stale; verified actual source uses logout(email), clears auth and prefills email. Backend role authority checked by [test_jwt_role_is_never_authority]\(tests/test_reliability.py#L57). No instant role token is issued.\
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\## PHASE 8: AUTOMATED TEST EXECUTION\
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\- **QA139 — PASS** — Unit tests\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
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\- **QA140 — PASS** — API/integration tests\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
\
\- **QA141 — PASS** — Database migration tests\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
\
\- **QA142 — NOT APPLICABLE** — Frontend lint and build\
  Evidence: Plain JavaScript/CSS served by FastAPI; no configured frontend lint/build command, package.json or bundler. Executed node --check and5 locale tests instead; no frontend build success is claimed.\
\
\- **QA143 — PASS** — Backend lint and type checks, if configured\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
\
\- **QA144 — PASS** — End-to-end tests, if configured\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
\
\- **QA145 — PASS** — Dependency/security audit, if available\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
\
\- **QA146 — PASS** — Docker build and service health checks, if safe and configured\
  Evidence: [QA_TEST_RESULTS.md]\(QA_TEST_RESULTS.md) records executed commands, exit codes, scope and limitations; no external production systems used.\
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\## PHASE 9: REQUIRED ACCEPTANCE TESTS\

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\### Acceptance\
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\- **QA147 — PASS** — A valid signed gateway request is accepted.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py
#L60)\
\
\- **QA148 — PASS** — An invalid signature is rejected.\   Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py
#L60)\
\
\- **QA149 — PASS** — An expired timestamp is rejected.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py
#L60)\
\
\- **QA150 — PASS** — A reused nonce is rejected.\
  Evidence:
[test_valid_tampered_old_unknown_retired_and_replayed_signatures]\(tests/test_gateway_service.py
#L60); [test_gateway_restart_recovers_jobs_and_nonces]\(tests/test_gateway_service.py#L140)\
\
\- **QA151 — PASS** — Repeating a document request with the same idempotency key creates
only one document.\
  Evidence:
[test_idempotency_persists_and_rejects_changed_body]\(tests/test_gateway_service.py#L106);
[test_concurrent_same_key_creates_one_job_and_document]\(tests/test_gateway_service.py#L124)\
\
\- **QA152 — PASS** — Manager approval succeeds while the gateway is offline.\
  Evidence:
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109)
; [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA153 — PASS** — The outbox event remains pending and retries after recovery.\
  Evidence:
[test_reconciliation_recovers_lost_callback]\(tests/test_integration_gateway.py#L159);
[test_gateway_restart_recovers_jobs_and_nonces]\(tests/test_gateway_service.py#L140); [docker-
journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA154 — PASS** — Exhausted retries become DEAD and are visible to HR.\
  Evidence:
[test_approval_survives_gateway_down_retries_then_dead]\(tests/test_integration_gateway.py#L109)
;
[test_validation_failure_not_retried_and_hr_queue_shows_it]\(tests/test_integration_gateway.py#L14
5)\
\
\- **QA155 — PASS** — A valid callback updates the external reference and integration status.\
  Evidence:
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79); [docker-
journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\

\- **QA156 — PASS** — Repeated callback processing does not duplicate notifications.\
  Evidence:
[test_signed_full_journey_notifies_exactly_once]\(tests/test_integration_gateway.py#L79); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA157 — PASS** — Employee tokens cannot invoke privileged callbacks or manager/HR operations.\
  Evidence:
[test_bearer_cannot_callback_and_tampered_or_replay_callback_rejected]\(tests/test_integration_gateway.py#L174);
[test_all_gateway_endpoints_reject_employee_bearer]\(tests/test_gateway_service.py#L94);
[test_all_administrative_operations_check_role]\(tests/test_api_matrix.py#L183)\
\
\- **QA158 — PASS** — Profile and last-known HR snapshot remain available while the gateway is down.\
  Evidence:
[test_refresh_is_hr_only_rate_limited_and_stale_snapshot_survives_outage]\(tests/test_integration_gateway.py#L249); [docker-journey.json]\(docs/evidence/qa-2026-10-17/docker-journey.json)\
\
\- **QA159 — PASS** — A manager cannot approve requests outside their authorized scope.\
  Evidence: [test_scope_and_hr_queue]\(tests/test_reliability.py#L132);
[test_manager_opens_team_member_card_but_not_others]\(tests/test_employee_directory.py#L30)\
\
\- **QA160 — PASS** — A failed transaction leaves no partial approval/outbox state.\
  Evidence:
[test_failed_approval_rolls_back_outbox_and_all_business_effects]\(tests/test_qa_acceptance.py#L89)
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\
\- **QA161 — PASS** — Course progress persists after reload.\
  Evidence:
[test_explicit_enrollment_progress_and_persistence]\(tests/test_reliability.py#L283); [relogin-persistence.json]\(docs/evidence/qa-2026-10-17/relogin-persistence.json)\
\
\- **QA162 — PASS** — AI Jana returns a source-backed answer or a safe fallback.\